

# **MUTUAL FUND ANALYTICS CAPSTONE PROJECT**

## **Internship Final Report**

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**Internship Role:** Data Analyst Intern

**Organization:** Bluestock Fintech

**Branch:** Artificial Intelligence & Data Science

**College:** PDEA College of Engineering, Manjari Budruk, Pune

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## Project Overview

The **Mutual Fund Analytics Capstone Project** was successfully completed as part of the **Bluestock Fintech Data Analyst Internship**. The project aimed to develop a complete data analytics solution for Indian mutual fund datasets using modern data analysis and business intelligence tools. The workflow included data collection, cleaning, transformation, database creation, exploratory data analysis, financial performance evaluation, advanced analytics, and interactive dashboard development.

The project focused on converting raw financial datasets into meaningful insights that assist investors in understanding fund performance, investment trends, portfolio risk, and investor behaviour through effective visualizations and analytical techniques.

## Project Objectives

- Design and implement an automated ETL pipeline for processing multiple mutual fund datasets.
- Clean and preprocess financial data by removing duplicate records, handling missing values, and standardizing date formats.
- Develop a SQLite database for efficient storage and management of processed datasets.
- Perform Exploratory Data Analysis (EDA) to identify investment trends and market behaviour.
- Calculate important financial performance indicators such as Sharpe Ratio, Alpha, Beta, CAGR, Historical VaR, and CVaR.
- Conduct advanced analytics including Investor Cohort Analysis, SIP Continuity Analysis, Sector HHI Concentration Analysis, and a Mutual Fund Recommendation System.
- Build an interactive Power BI dashboard to visualize key performance metrics and support data-driven investment decisions.

## Tools & Technologies Used

**Programming Language:** Python

**Libraries:** Pandas, NumPy, Matplotlib, Seaborn, Plotly

**Database:** SQLite

**Query Language:** SQL

**Business Intelligence Tool:** Power BI

**Development Environment:** Jupyter Notebook, Visual Studio Code

## Project Work Completed

During the internship, an automated ETL pipeline was developed to process multiple CSV datasets containing mutual fund information. The pipeline successfully extracted raw data, removed duplicate entries, handled missing values, standardized date formats, and generated cleaned datasets for further analysis.

A relational SQLite database was created to store the processed data efficiently. SQL queries were executed to validate the dataset and perform analytical operations.

Comprehensive Exploratory Data Analysis (EDA) was conducted to examine Net Asset Value (NAV) trends, Assets Under Management (AUM) growth, SIP inflow trends, category-wise investment patterns, investor demographics, geographical distribution, folio growth, and fund return correlations.

Advanced financial analytics were performed by calculating Sharpe Ratio, Alpha, Beta, Historical Value at Risk (VaR), Conditional Value at Risk (CVaR), Rolling Sharpe Ratio, Sector Concentration using the Herfindahl-Hirschman Index (HHI), Investor Cohort Analysis, SIP Continuity Analysis, and a rule-based Mutual Fund Recommendation System based on investor risk appetite.

A professional four-page Power BI dashboard was designed with interactive visualizations, KPI cards, slicers, charts, and performance metrics to provide an intuitive overview of mutual fund performance and investor insights.

## Key Findings :

- Mutual Fund SIP inflows demonstrated a consistent upward trend between 2022 and 2025, indicating increasing investor participation.
- Equity-oriented mutual funds recorded the highest net inflows among all fund categories.
- Several mutual fund schemes achieved superior risk-adjusted returns based on Sharpe Ratio analysis.
- Historical VaR and CVaR analysis helped identify funds with comparatively higher downside risk.
- Investor participation increased significantly across multiple age groups and geographical regions.
- Interactive dashboards simplified the analysis of fund performance, risk, and investment trends through dynamic filtering and visualization.

## Conclusion :

The **Mutual Fund Analytics Capstone Project** successfully demonstrated an end-to-end data analytics workflow covering ETL, database management, exploratory data analysis, financial performance measurement, advanced analytics, and business intelligence reporting.

This internship significantly enhanced practical knowledge of Python, SQL, SQLite, Jupyter Notebook, and Power BI while improving analytical thinking, problem-solving abilities, and data visualization skills. The developed solution provides valuable insights into mutual fund performance and supports informed investment decision-making using data-driven techniques.